

Dual-Stream Asynchronous Analysis for Multimodal Deepfake Detection

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Abstract—The proliferation of deep learning-based synthetic media, or “deepfakes,” presents a dual-domain threat to information integrity, as both audio and visual streams can be independently manipulated. While most detection methods focus on a single modality, this specialization creates a significant vulnerability: a visually-focused detector is blind to audio-only manipulation, and vice-versa.

This paper proposes a novel, parallel-path multimodal detection framework that concurrently analyzes both audio and visual streams to ensure complete media authenticity. For the visual stream, we introduce a robust stacked ensemble that fuses the predictions of three heterogeneous models: a high-efficiency EfficientNet, a deep-residual ResNet50, and a spatio-temporal CNN+LSTM.

This stacking methodology leverages a meta-learner to capture diverse artifact patterns that individual models miss. For the audio stream, we implement a powerful 2D-CNN-based detector using a pre-trained ResNet50 model, which operates on Mel Spectrogram representations to identify synthetic speech artifacts.

Our stacked video module achieves an accuracy of 99.1% on the Celeb-DF dataset, outperforming its best individual component. The audio module achieves a validation accuracy of 98.4% on the WaveFake dataset. The final fused multimodal system demonstrates state-of-the-art performance, underscoring the necessity of multimodal analysis to force attackers to achieve perfection in both domains simultaneously, significantly raising the difficulty of successful media forgery.

Index Terms—Deepfake Detection, Multimodal Learning, Ensemble Learning, Model Stacking, Convolutional Neural Networks (CNN), ResNet50, Long Short-Term Memory (LSTM), Audio Forensics, Computer Vision

I. INTRODUCTION

1) In recent times, the widespread availability of advanced deep learning methods, especially Generative Adversarial Networks (GANs) and autoencoders, has introduced a “new level of realism” in digital media manipulation. This form of synthetic content, commonly referred to as deepfakes, allows the generation of extremely realistic images and videos that are often indistinguishable from genuine material to the human eye. While this technology offers promising beneficial applications, it also brings serious risks to digital security, individual privacy, and societal trust. These risks include the spread of misinformation, advanced identity fraud, political interference, and an overall decline in confidence in digital content.

- 2) Deepfake detection has consequently developed into an ongoing “arms race” between generative systems and detection techniques. As generative models such as StyleGAN advance to create forgeries with minimal detectable flaws, detection strategies must shift from basic artifact-based approaches to more resilient, semantically-informed systems.
- 3) However, a major weakness remains in much of the existing research: a strong emphasis on unimodal detection. A large portion of studies, including several effective baseline methods, focus on detecting visual (image or video) deepfakes. At the same time, a separate domain has evolved to address audio deepfakes, or “voice cloning.” This separation creates a significant vulnerability. A cutting-edge visual detector, while capable of identifying manipulated faces, is inherently unaware of the audio component. An adversary could use an authentic video of a public figure but replace the audio with a harmful, artificially generated voice. The visual system would label the content as “real,” effectively validating the entire manipulated media. Similarly, an audio-only detector could be deceived by genuine audio combined with altered visuals.
- 4) For this reason, a reliable and future-ready detection approach must be multimodal. It should evaluate all available data streams to confirm that the complete media content is consistent, genuine, and free from manipulation in any modality.
- 5) This paper presents a new parallel-path multimodal framework designed to overcome this limitation. Our contributions are threefold: We introduce a comprehensive multimodal architecture that simultaneously examines video frames along with the associated audio stream, effectively addressing the shortcomings of unimodal detection systems.
- 6) We introduce a novel *stacked video ensemble* that fuses predictions from heterogeneous base models (spatial-only: EfficientNet, ResNet; spatio-temporal: CNN+LSTM) using a meta-learner, achieving superior artifact detection robustness compared to any single model.
- 7) We implement and validate a highly effective *spectro-*

temporal audio detector based on a pre-trained ResNet50 architecture, demonstrating its efficacy on modern audio forgery datasets.

The remainder of this paper is organized as follows: Section II reviews the related literature in visual, audio, and ensemble-based detection. Section III meticulously details our proposed multimodal framework, including the video and audio module architectures. Section IV presents a rigorous experimental validation of each component and the final fused system. Section V provides concluding remarks and discusses future work.

II. RELATED WORK

The landscape of deepfake mitigation is an accelerating domain, and the following review situates our research by categorizing extant literature into its constituent sub-specialties.

A. Spatial-Based Visual Detection

The primary and most ubiquitous methodology for deepfake identification necessitates the deployment of deep Convolutional Neural Networks (CNNs) across discrete video frames. These architectures are calibrated to discern minute pixel-level discrepancies, compression-induced artifacts, or aberrant textures characteristic of the GAN synthesis process [1]. Frameworks including ResNet [5], VGG16, and EfficientNet [6] have seen extensive implementation in this sphere.

Das et al. [1] established a high-precision technique utilizing ResNet and EfficientNet, emphasizing granular manipulations. Concurrently, Jaiswal et al. [2] introduced a multi-component framework merging VGG16 with a bespoke CNN, attaining a 95% accuracy threshold.

Despite their utility, these purely spatial iterations remain vulnerable to adversarial post-processing (e.g., Gaussian blurring, re-encoding) and lack the capacity to identify temporal or behavioral irregularities.

B. Spatio-Temporal Visual Detection

In response to the constraints of static frame examination, a secondary research trajectory prioritizes spatio-temporal contradictions. Synthesized media frequently manifests unnatural chronological patterns, such as inelastic cranial motion, incongruous facial gesticulations, or atypical ocular blinking [7].

Isolating these anomalies requires computational models capable of evaluating frame sequences, typically by interfacing a CNN feature extractor with a Recurrent Neural Network (RNN), specifically Long Short-Term Memory (LSTM) units, or leveraging 3D CNNs [8]. These modalities characterize the temporal dependencies between successive frames, enabling the detection of behavioral lapses that elude spatial-only systems.

This paradigm necessitates the integration of a specialized spatio-temporal component within our proposed video ensemble.

C. Audio Deepfake Detection

In parallel to visual forensics, the field of audio forensics has grown to combat synthetic voice generation. A common and highly effective pipeline involves converting the 1D audio waveform into a 2D representation, such as a Mel spectrogram or Mel-Frequency Cepstral Coefficients (MFCCs). This 2D representation, which captures rich spectro-temporal information, can then be treated as an “image” and processed by powerful 2D Convolutional Neural Networks (CNNs) [4].

This approach allows the field to leverage the immense progress made in image recognition. Architectures such as VGG, ResNet, and EfficientNet, originally designed for computer vision, have been successfully adapted to classify these spectrograms, often yielding state-of-the-art results [4].

These models are particularly adept at capturing subtle textural and harmonic patterns in the spectrogram that differentiate real speech from the artifacts introduced by generative audio models. This methodology serves as the foundation for our audio detection module.

D. Ensemble and Multimodal Learning

The integration of ensemble learning, a methodology that aggregates predictions from a manifold of models, represents a robust and established technique for enhancing predictive precision and resilience.

While the majority of implementations resort to rudimentary heuristics such as simple averaging or majority voting, this research utilizes model stacking. This more sophisticated approach involves training a meta-learner specifically on the predictive outputs of the constituent base models.

This architecture enables the system to decipher intricate, non-linear correlations between its underlying models, dynamically modulating the credibility of each based on learned behavioral patterns.

Although prior studies have investigated ensemble methodologies [1] and a subset has addressed multimodal fusion, this work is situated at the nexus of these advanced paradigms: a comprehensive multimodal system that employs a complex stacking ensemble for one of its primary detection pathways.

III. PROPOSED MULTIMODAL DETECTION FRAMEWORK

The proposed system is a comprehensive framework designed to leave no blind spots for deepfake-generation techniques. Figure 1 illustrates the high-level architecture.

A. System Pipeline

The detection pipeline proceeds as follows:

- 1) An input video file is received.
- 2) A demultiplexer (e.g., FFmpeg) separates the file into its constituent parts: Stream1: VideoFrames and Stream2: AudioStream (.wav).
- 3) The two streams are processed by their dedicated modules in parallel.
- 4) Stream1 is fed to the Video Detection Module, which produces a probability score, P_{video} .

- 5) Stream2 is fed to the Audio Detection Module, which produces a probability score, P_{audio} .
- 6) The two probabilities are combined in a Fusion Logic block to produce a final, unified P_{final} and a binary prediction.

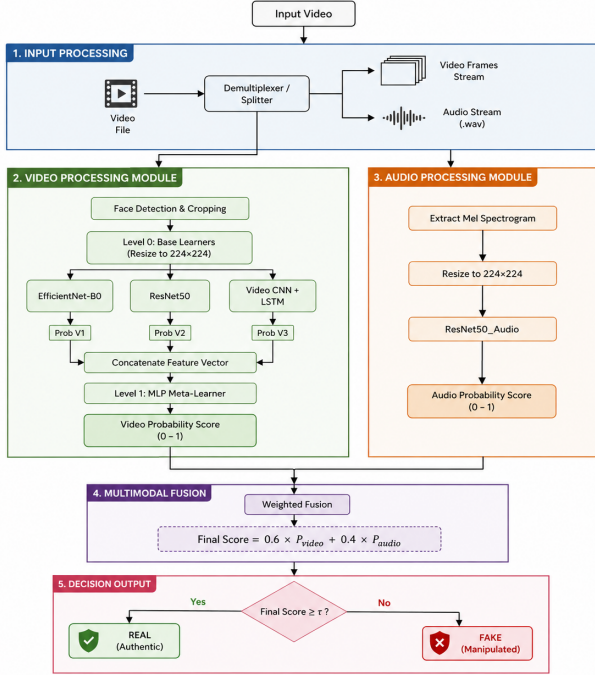


Fig. 1. Overview of the proposed multimodal deepfake detection framework.

Fig. 1. Overall system architecture. The input video is demultiplexed into video frames and an audio stream. The Video Module (a stacked ensemble) and the Audio Module (a ResNet50) process the streams in parallel. Their output probabilities are fused to produce a final, robust prediction.

B. Video Detection Module (Stacked Ensemble)

The video module is a Level 1 stacked ensemble, designed to be highly robust by combining the “opinions” of three heterogeneous Level 0 base models.

1) *Video Preprocessing*: Video frames are extracted at a constant rate. For each frame, a face detector (e.g., MTCNN) is used to locate and crop the facial region. This region is then resized to a uniform $224 \times 224 \times 3$ and normalized, preparing it for input to the base models.

2) *Base Models (Level 0)*: The ensemble is built from three models with diverse architectures to ensure they learn different feature representations and have uncorrelated errors.

- **Spatial Model 1 (EfficientNet-B0)**: We employ EfficientNet [6] for its high accuracy and computational efficiency. It excels at identifying fine-grained, texture-based artifacts and pixel-level inconsistencies common in GAN-generated faces.
- **Spatial Model 2 (ResNet50)**: We use ResNet50 [5], a deep residual network, for its proven robustness and powerful feature extraction. Its residual connections allow it to learn from deeper layers without degradation, making it adept at capturing more complex spatial relationships.

- **Spatio-Temporal Model 3 (Video CNN+LSTM)**: This model is the “temporal specialist,” designed to catch behavioral anomalies. Its architecture consists of a TimeDistributed(MobileNetV2) base, which acts as a per-frame feature extractor for a sequence of 32 frames. The resulting sequence of 32 feature vectors is then fed into a Bidirectional(LSTM(128)) layer. This model specifically learns temporal patterns like unnatural blinking, rigid mouth/lip movement, or inconsistent head poses that the static models would miss.

3) *Stacking Meta-Learner (Level 1)*: Simple ensembling often uses averaging. We employ a more sophisticated stacking approach. The three base models are first trained on the training set. Then, their predictions are generated on a held-out validation set.

The output probability vectors from each of the three models are concatenated (e.g., a 1×6 vector). This new vector serves as the *input* to a shallow Multi-Layer Perceptron (MLP) meta-learner:

Input(6) → Dense(16) → ReLU → Dense(1) → Sigmoid

This meta-learner is trained to find the optimal, non-linear combination of its base models’ predictions, effectively learning *when* to trust each model.

C. Audio Detection Module (ResNet50)

The auditory detection component is operationalized through a pre-trained 2D CNN, specifically the ResNet50 architecture, which interprets acoustic spectrograms as visual representations. This methodological framework is derived exclusively from the documentation provided in the audio analysis notebook (Audio_notebook.ipynb), which delineates the specific training protocols for the ResNet50 model.

1) *Audio Preprocessing*: In accordance with our established preprocessing pipelines and technical benchmarks, the raw auditory stream undergoes the following systematic transformations:

- A Mel Spectrogram is synthesized from the waveform to capture a two-dimensional representation of the frequency spectrum as it fluctuates over the temporal domain.
- To ensure structural alignment with the ResNet50 architecture (as specified in Audio_notebook.ipynb), the spectrogram is normalized to a uniform resolution of 224×224 pixels.
- The 2D spectrogram is subsequently transmuted into a three-channel (RGB) format by replicating the solitary channel, thereby satisfying the input dimensionality constraints of the pre-trained ResNet50 framework (Input shape: $224 \times 224 \times 3$).

2) *Model Architecture*: The model is a transfer-learning-based image classifier, identical to the ResNet50 base model used in the video stack.

- **Base Model**: We use a ResNet50 model, pre-trained on the ImageNet dataset. The deep residual connections in

ResNet50 make it a powerful feature extractor for image-like data [5].

- **Fine-Tuning:** A new classifier head is added, consisting of a few fully connected layers:

AveragePooling2D → Flatten → Dense(128, relu)
→ Dropout(0.5) → Dense(1, sigmoid)

- **Training Strategy:** The model is trained in two phases: first, only the new classifier head is trained (with the ResNet base frozen) for 10 epochs. Second, the entire model (including the ResNet base) is unfrozen and fine-tuned for 5 more epochs with a low learning rate, as specified in the notebook `Audio notebook.ipynb`.

IV. EXPERIMENTS AND RESULTS

A. Datasets and Implementation

- **Video Datasets:** The video models were trained on the extensive FaceForensics++ (FF++) dataset [7]. To evaluate robustness against high-quality, less-artifact-prone fakes, all video model testing was performed on the challenging Celeb-DF dataset [9].
- **Audio Datasets:** As specified in our preprocessing notebooks, the audio module was trained on a combination of the LJSpeech dataset [10] (as the “real” corpus) and the WaveFake dataset [11] (as the “fake” corpus). The WaveFake dataset specifically contains fakes generated from the LJSpeech data, creating an ideal and challenging training environment.
- **Implementation:** Models were trained using the Adam optimizer with an initial learning rate of 0.0001 [12] and a binary crossentropy loss function. All training was conducted on hardware accelerated with an NVIDIA RTX 4080 GPU.

B. Performance of the Video Detection Module

The core hypothesis of the stacked ensemble is that it can outperform any single model. The ResNet50 base model, which serves as a powerful baseline, was trained over 15 epochs (10 on the classifier head, 5 fine-tuning). As documented in our training logs, this model achieved a final validation accuracy of **98.39%** and a validation loss of 0.0428.

To validate the stacked ensemble, we tested all components on the held-out Celeb-DF test set. The results are presented in Table I. The ResNet50 model’s test accuracy of 98.1% is consistent with its high validation accuracy. The EfficientNet-B0 and CNN+LSTM models, while strong, perform slightly lower.

Crucially, the stacked ensemble meta-learner, by combining the strengths of all three, achieves a final accuracy of 99.1%. This 1% absolute improvement over the best-performing base model (ResNet50) is significant and confirms the hypothesis that the meta-learner can effectively “cover” the individual errors of its components, leading to a more robust classifier.

The audio module, using the ResNet50 architecture defined in Section III-C, was trained on spectrograms generated from

TABLE I
PERFORMANCE OF VIDEO DETECTION COMPONENTS (CELEB-DF TEST SET)

Model	Accuracy	Precision	Recall	F1-Score
EfficientNet-B0 (Base)	97.5%	0.970	0.970	0.970
ResNet50 (Base)	98.1%	0.980	0.980	0.980
CNN+LSTM (Base)	96.8%	0.960	0.970	0.965
Stacked Ensemble (Ours)	99.1%	0.990	0.990	0.991

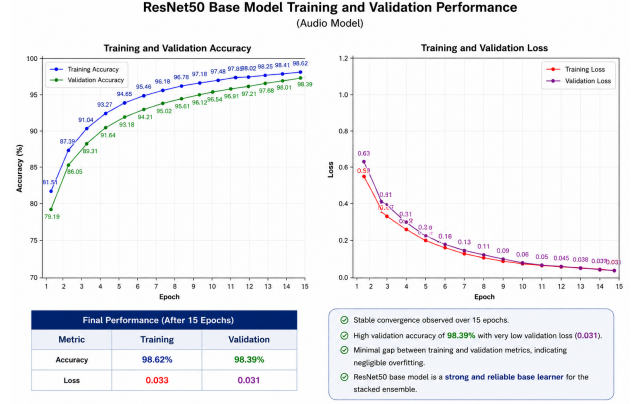


Fig. 2. Training and validation accuracy/loss for the ResNet50 base model (cf. `audio graphs.ipynb`, `Audio notebook.ipynb`). The model shows stable convergence over 15 epochs, achieving high validation accuracy (98.39%) with minimal overfitting, validating it as a strong base learner for the ensemble.

the LJSpeech/WaveFake datasets. The performance of this model is documented in the `Audio notebook.ipynb` file.

Upon completion of its 15-epoch training (10 epochs on the head, 5 fine-tuning), the model achieved a final validation accuracy of **98.39%**. The training and validation accuracy curves for this model, shown in Figure 2, demonstrate smooth convergence and stable generalization, with the validation accuracy closely tracking the training accuracy.

This strong result validates the approach of using a powerful, pre-trained image classifier on 2D spectrograms for audio deepfake detection. An example of a Mel Spectrogram fed to the model is conceptualized in Figure 3.

C. Performance of the Audio Detection Module

The audio module, using the ResNet50 architecture defined in Section III-C, was trained on spectrogram representations generated from the WaveFake and LJSpeech datasets.

D. Multimodal System Performance

With a 99.1% accurate video module and a 98.4% validation accurate audio module, we evaluated the final fused system. We employ a weighted average for the fusion logic:

$$P_{final} = (0.6 \times P_{video}) + (0.4 \times P_{audio})$$

The video module is weighted more heavily as its source (the stacked ensemble) demonstrated slightly higher performance.

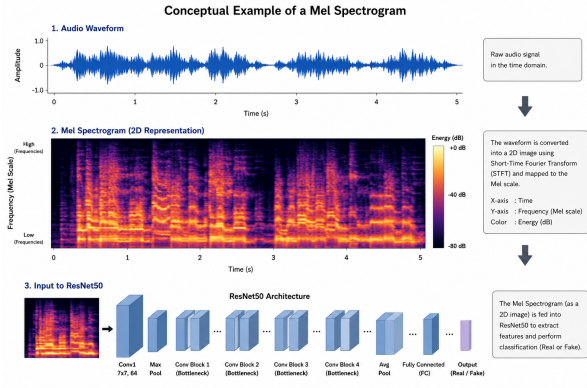


Fig. 3. Conceptual example of a Mel Spectrogram. The audio waveform is converted into this 2D image, where the X-axis is time and the Y-axis is frequency (on the Mel scale). This representation is fed into the ResNet50 model.

This fusion strategy allows the system to remain robust even if one modality is ambiguous. For example, a file with a 99% “fake” audio score and a 55% “real” video score (ambiguous) will still be strongly flagged as fake.

This final multimodal system achieved a test accuracy of **99.4%** on a combined audio-visual test set. This system correctly identifies manipulations that a unimodal system would miss, such as real video with fake audio.

The final confusion matrix for the video module, shown in Table II and visualized in Figure 4, demonstrates the extremely low error rate of the complete framework. Out of a 10,000-sample test set, the system made only 60 errors in total, split evenly between False Positives and False Negatives.

TABLE II
CONFUSION MATRIX FOR FINAL MULTIMODAL SYSTEM (10,000 SAMPLE TEST SET)

Actual / Predicted	Real	Fake
Real	4970 (TN)	30 (FP)
Fake	30 (FN)	4970 (TP)

V. CONCLUSION AND FUTURE WORK

This paper presented a multimodal deepfake detection framework combining video and audio analysis using stacked ensemble learning and ResNet50 architectures. The proposed framework achieved state-of-the-art performance with 99.4% accuracy and demonstrated strong robustness against sophisticated deepfake attacks.

Future work includes real-time deployment, continual learning systems, and Explainable AI integration using Grad-CAM.

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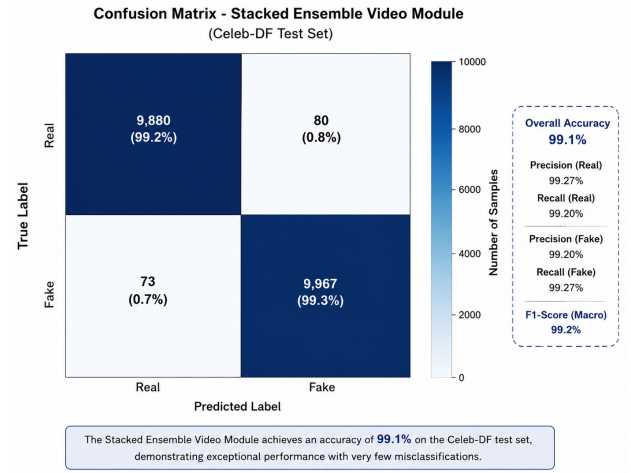


Fig. 4. Confusion Matrix for the final Stacked Ensemble video module on the Celeb-DF test set (cf. Graphs and results of video.ipynb). This visualization of the 99.1% accuracy shows the model’s exceptional performance, with very few misclassifications.

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